Prostate cancer

Managed on two axes — **extent** (localized → PSA-only → metastatic) crossed with **castration status** (sensitive vs resistant, testosterone <50 ng/dL). **Test every metastatic patient for germline + somatic HRR** — it now changes therapy in mHSPC as well as mCRPC

Compiled from the Stanford Oncology Handbook 2025, Prostate Cancer (Chien J, Parikh D, Srinivas S) · NCCN · primary trial reports verified against PubMed, September 2026.

SCREENING & DIAGNOSIS

- **USPSTF (Grade C): shared decision on PSA at 55–69**; not recommended ≥70. Consider starting at 40–45 for Black men and BRCA2 carriers
- **Pre-biopsy multiparametric MRI** (PI-RADS) → targeted biopsy. **PRECISION:** clinically significant cancer **38% vs 26%** vs standard TRUS biopsy, with fewer insignificant cancers, and 28% of the MRI arm avoided biopsy
- Risk inputs: **cT stage, Grade Group, PSA**, % positive cores, PSA density. Tissue genomic classifiers (Decipher, Prolaris) can refine AS vs treatment
- **Staging imaging by risk:** none for very low / low / favorable intermediate · soft tissue ± bone for unfavorable intermediate · **bone + soft tissue for high / very high**
- **PSMA PET** (Ga-68 PSMA-11, F-18 piflufolastat) beats CT + bone scan. **proPSMA:** accuracy **92% vs 65%**, fewer equivocal reads. Preferred for BCR restaging and required before Lu-PSMA

GRADE GROUP & NCCN RISK (LOCALIZED)

Grade Group	GG1 ≤3+3 · GG2 **3+4** · GG3 **4+3** · GG4 = 8 · GG5 = 9–10
Very low	**All of:** T1c, GG1, PSA <10, <3 cores positive (≤50% each), PSA density <0.15
Low	cT1–T2a + GG1 + PSA <10, not very low
Favorable interm.	**1 IRF** + GG1–2 + <50% cores positive
Unfavorable interm.	**2–3 IRFs, or GG3, or ≥50% cores** positive
High	**One of:** cT3–T4, GG4–5, PSA >20
Very high	**≥2 of:** cT3–T4, GG4–5, PSA >40

IRF (intermediate risk factor) = cT2b–T2c, GG2–3, or PSA 10–20. Stage IVA = N1 M0; IVB = M1 (M1a non-regional nodes, M1b bone, M1c other ± bone).

LOCALIZED DISEASE

- **Active surveillance** for very low / low and selected favorable intermediate: PSA q6 mo, repeat MRI ± biopsy; treat on upgrading
- **Definitive: radical prostatectomy or EBRT ± brachytherapy boost** — similar cancer control, different toxicity (RP: incontinence, ED; RT: bowel/bladder)
- **ADT with RT scales with risk:** unfavorable intermediate **4–6 mo**; high / very high **18–36 mo** (~12 mo if brachy boost). **Very high or N1: add abiraterone ×2 yr** (STAMPEDE)
- **2026 IPD meta-analysis** (13 RCTs, n=10,266): benefit **plateaus beyond ~9–12 mo** while non-cancer death keeps rising — argues for the **short end** of each range

TREATMENT PATHWAY BY DISEASE STATE

Localized (M0, castration-sensitive)

Risk-adapted local therapy
low: **active surveillance**
interm./high: RP, or EBRT + ADT 4–6 or 18–36 mo
very high / N1: EBRT + ADT + **abiraterone 2 yr**

Biochemical recurrence (PSA only)

Restage with PSMA PET, then
post-RP: **salvage RT ± ADT** (RTOG 9601, SPPORT)
high-risk BCR (PSADT ≤9 mo): **enzalutamide + leuprolide** (EMBARK — OS benefit, HR 0.60)

nmCRPC (castrate T, rising PSA, M0)

PSADT ≤10 mo → add an ARPI
apalutamide, darolutamide or enzalutamide
MFS gain ~2 yr, OS HR 0.69–0.78
(SPARTAN / ARAMIS / PROSPER)

mHSPC (metastatic, castration-sensitive)

ADT alone is never enough
doublet: ADT + ARPI (abi, apa, daro, enza)
high volume / de novo: triplet + docetaxel ×6
(PEACE-1, ARASENS); low volume: + RT to primary

Biomarker add-ons (2025–26)
BRCA2: niraparib + abiraterone (AMPLITUDE)
PTEN-deficient: capivasertib + abi (CAPItello-281)
PSMA+: Lu-PSMA + ARPI (PSMAddition)

mCRPC

Rotate mechanisms — test HRR first
never ARPI → ARPI (CARD: cabazitaxel wins)
BRCA/HRR: PARPi (+ ARPI if ARPI-naive)
docetaxel → cabazitaxel · **PSMA+**: Lu-PSMA
bone-only symptomatic: radium-223 · MSI-H: pembro

METASTATIC RISK DEFINITIONS

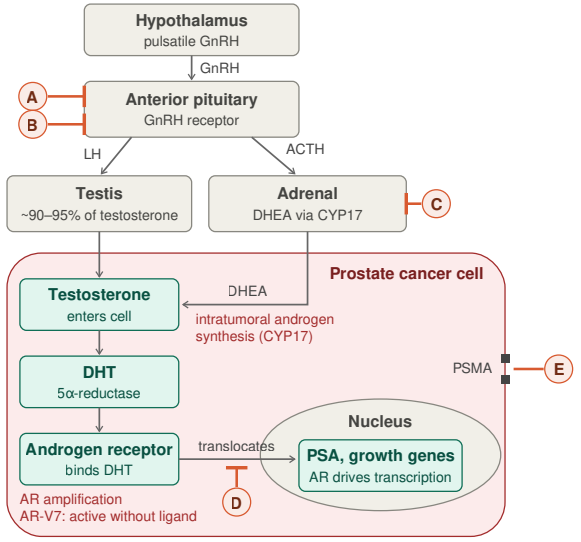
CHAARTED high volume	**Visceral mets and/or ≥4 bone lesions with ≥1 beyond the spine/pelvis**
LATITUDE high risk	**≥2 of:** Gleason ≥8, ≥3 bone lesions, measurable visceral mets
Castration resistance	Progression (PSA, radiographic) with **testosterone <50 ng/dL**
Timing	**De novo** worse than metachronous; favors triplet

Volume is judged on CT + bone scan; unvalidated for PSMA-only disease.

GERMLINE & SOMATIC TESTING

Germline	**All metastatic or N1**, high/very-high risk, high-risk BCR, Ashkenazi, family history (breast, ovarian, pancreatic, prostate, Lynch); consider if ≤55 or intraductal
Frequency *metastatic*	Germline DNA-repair mutation **11.8%**: **BRCA2 5.3%**, CHEK2 1.9%, ATM 1.6%, BRCA1 0.9%, regardless of age / family history
Somatic	**HRR panel** (tissue or ctDNA) in all metastatic. **MSI-H/dMMR + TMB** in mCRPC. **PTEN IHC** (companion test, 2026) in mHSPC
Why it matters	BRCA2 → PARP benefit is largest, and **cascade testing** of relatives. ATM/CDK12/CHEK2 respond poorly to PARPi

PATHOPHYSIOLOGY



(A) leuprolide: GnRH agonist (initial flare) (B) degarelix, relugolix: antagonist, no flare
(C) abiraterone: CYP17 inhibitor; + prednisone for ACTH-driven mineralocorticoid excess
(D) enzalutamide, apalutamide, darolutamide: AR pathway inhibitors (binding + translocation)
(E) Lu-177-PSMA-617: PSMA radioligand

ADT is continued in CRPC: AR signaling persists (amplification, AR-V7, tumor androgens).
Red text = castration-resistance mechanisms.

Prostate cancer

LOCALIZED & HIGH-RISK M0 — RT + ADT

Trial	Design → result
RTOG 9408	RT ± **4 mo** ADT, mostly intermediate risk → 10-yr OS **62% vs 57%**; benefit confined to intermediate risk
EORTC 22863	RT ± **3 yr** ADT, locally advanced → 10-yr OS **58% vs 40%** (HR 0.60)
EORTC 22961	**6 mo vs 3 yr** → 6 mo **not non-inferior** (5-yr OS 81% vs 85%)
RTOG 9202 · DART	Short vs long → long course wins mainly in **Gleason 8–10** / **high risk** (9202 GS 8–10 10-yr OS 45% vs 32%)
PCS IV	**18 vs 36 mo**, high risk → 10-yr OS 62% vs 62%; 18 mo sufficient, better QoL
Zaorsky 2026 IPD	13 RCTs → optimal ADT: **0 mo** for 1 IRF, **6 mo** ≥2 IRFs, **12 mo** high risk; gains flatten after ~9–12 mo
STAMPEDE (abi M0)	High-risk N0 or N1: ADT 3 yr ± **abiraterone 2 yr** → MFS HR **0.53**, 6-yr MFS 82% vs 69%, OS HR 0.60. **Adding enzalutamide added nothing**
PATCH (2026)	**Transdermal estradiol vs LHRH**, locally advanced M0 → non-inferior MFS (HR 0.96); fractures 5% vs 11%, hot flashes far fewer; **gynecomastia 85%**

BIOCHEMICAL RECURRENCE & NMCRPC

Trial	Design → result
RTOG 9601	Salvage RT ± **bicalutamide 150 mg ×24 mo** → 12-yr OS **76% vs 71%** (HR 0.77); gynecomastia 70%
SPPORT	Salvage bed RT vs + 4–6 mo ADT vs + ADT + pelvic nodes → 5-yr freedom from progression **71%** / **81%** / **87%**
EMBARK	High-risk BCR (PSADT ≤9 mo): enza + leuprolide vs leuprolide vs enza mono → 5-yr MFS **87%** / **71%** / **80%**. **Final OS (2025): combination HR 0.60** (8-yr 79% vs 70%); **enzalutamide monotherapy HR 0.83, NOT significant**
PRESTO	Post-RP BCR, **52 wk** ADT ± apalutamide (± abi) → PSA-PFS 24.9 vs 20.3 mo (HR 0.52); no OS data
SPARTAN	nmCRPC, PSADT ≤10 mo: **apalutamide** → MFS 40.5 vs 16.2 mo; OS **73.9 vs 59.9** (HR 0.78)
PROSPER	**Enzalutamide** → MFS 36.6 vs 14.7 mo; OS **67.0 vs 56.3** (HR 0.73)
ARAMIS	**Darolutamide** → MFS 40.4 vs 18.4 mo; OS HR **0.69** (3-yr 83% vs 77%); least CNS toxicity

EMBARK is the reason to use the **combination**, not enzalutamide alone, when survival is the goal; monotherapy is a compromise for a patient avoiding castration.

MHSPC — DOUBLET AND TRIPLET

Trial	Design → result
CHAARTED	ADT ± **docetaxel ×6** → OS 57.6 vs 44.0 mo (HR 0.61); **benefit concentrated in high volume** (49.2 vs 32.2)
LATITUDE	**High-risk** de novo: + **abiraterone** → 3-yr OS 66% vs 49% (HR 0.62)
TITAN	+ **apalutamide**, any volume → 24-mo OS 82% vs 74% (HR 0.67)
ARCHES	+ **enzalutamide** → rPFS HR 0.39; OS HR **0.66**
ENZAMET	+ enzalutamide vs + NSAA (docetaxel allowed) → 3-yr OS 80% vs 72% (HR 0.67)
ARANOTE	+ **darolutamide** (no docetaxel) → rPFS HR 0.54; **OS HR 0.81, not significant**. FDA June 2025
PEACE-1	De novo: ADT ± docetaxel ± **abiraterone** ± RT → high volume OS **5.1 vs 3.5 yr** with abi on top of docetaxel; RT to primary helps low volume
ARASENS	ADT + docetaxel ± **darolutamide** → OS HR **0.68**; little added toxicity

Triplet = high-volume, de novo, fit for docetaxel. **No trial randomized triplet vs ARPI doublet** — the triplet advantage over an ARPI doublet is inferred, not proven.

MHSPC — BIOMARKER-DIRECTED (2025–26)

Trial	Design → result · label
AMPLITUDE	HRR-altered (56% BRCA): **niraparib + AAP** vs AAP → rPFS HR **0.52 BRCA**, 0.63 ITT. Grade 3–4 AEs 75% vs 59%, anemia 29%. **FDA Dec 2025: BRCA2 only**
CAPItello-281	PTEN-deficient by IHC (~25%): **capivasertib + AAP** → rPFS 33.2 vs 25.7 mo (HR **0.81**). Diarrhea 52%, hyperglycemia 38%, rash 35%. **FDA June 2026**
PSMAddition	PSMA PET+: **Lu-PSMA ×6 + ADT + ARPI** vs ADT + ARPI → rPFS HR **0.72**. Dry mouth 46%. **FDA July 2026**

All three are rPFS wins with **immature or non-significant OS** and added toxicity — none yet displaces docetaxel triplet for high-volume disease.

CHOOSING MHSPC THERAPY

- **Low volume or metachronous:** ADT + ARPI doublet; add **RT to the prostate** for low-volume de novo disease
- **High volume, de novo, fit:** triplet — docetaxel + **darolutamide** (ARASENS) or **abiraterone** (PEACE-1)
- **Pick the ARPI by comorbidity:** darolutamide for falls, cognition, seizures, polypharmacy; avoid abiraterone with uncontrolled hypertension, diabetes, liver disease; enzalutamide/apalutamide are strong CYP inducers
- **Then check biomarkers:** BRCA2 → niraparib + AAP; PTEN-deficient → capivasertib + AAP; PSMA+ → Lu-PSMA add-on

MCRPC — PARP INHIBITORS: TRIAL VS FDA LABEL

Trial	Population → result	US label
PROfound	Post-ARPI, HRR-selected: **olaparib** vs ARPI switch → BRCA/ATM rPFS 7.4 vs 3.6 (HR 0.34); OS 19.1 vs 14.7	HRR-mutated, post-ARPI
TRITON2 / **3**	**Rucaparib**: single-arm BRCA ORR 44% → TRITON3 BRCA rPFS 11.2 vs 6.4 (HR 0.50); **ATM no benefit** (HR 0.95)	BRCA, post-ARPI; **regular Dec 2025**
PROpel	1L, **all-comers**: **olaparib + abiraterone** → rPFS HR 0.66 (BRCA 0.23); OS HR 0.81, NS	**BRCA only**
MAGNITUDE	1L, HRR-selected: **niraparib + AAP** → BRCA rPFS 16.6 vs 10.9 (HR 0.53); HRR-negative cohort stopped for futility	**BRCA only**
TALAPRO-2	1L, **all-comers**: **talazoparib + enzalutamide** → final OS 45.8 vs 37.0 (HR 0.80); **HRR-def OS HR 0.62**, BRCA 0.50, non-HRR 0.88 NS	**HRR-mutated**; all-comers rejected (ODAC 8–0, 2025)

Signal tracks **BRCA2 > BRCA1 > other HRR**; non-HRR benefit is small or absent. PARP + ARPI combinations are for **ARPI-naive** mCRPC; after an ARPI, use PARP monotherapy.

MCRPC — RADIOLIGAND, BONE, CHEMO

Trial	Design → result
VISION	PSMA+, post-ARPI + taxane: **Lu-PSMA-617** → OS 15.3 vs 11.3 mo (HR 0.62)
PSMAfore	Post-ARPI, **taxane-naive**: Lu-PSMA vs ARPI switch → rPFS 12.0 vs 5.6 (HR 0.41); **no OS gain** (84% crossover). FDA Mar 2025
ENZA-p *phase 2*	Enzalutamide ± Lu-PSMA, 1L mCRPC → OS 34 vs 26 mo (HR 0.55); not approved
ALSYMPCA	Symptomatic bone-only: **radium-223** → OS 14.9 vs 11.3 mo (HR 0.70)
ERA 223	**Abiraterone + radium** → fractures **29% vs 11%**, more deaths — unblinded early. **Never combine**
PEACE-3	**Enzalutamide + radium**, bone agent mandatory → rPFS 19.4 vs 16.4 (HR 0.69); final OS HR 0.76; fractures still 24% vs 13%
TAX 327 · TROPIC	Docetaxel q3w OS 18.9 vs 16.5 mo (HR 0.76) · post-docetaxel **cabazitaxel** 15.1 vs 12.7 (HR 0.70)
CARD	Post-docetaxel + 1 ARPI: **cabazitaxel vs 2nd ARPI** → rPFS 8.0 vs 3.7, OS 13.6 vs 11.0 (HR 0.64)

MCRPC — SEQUENCING

- **Most patients now reach mCRPC already ARPI-exposed** — a second ARPI is the weakest option (CARD)
- **ARPI-exposed: docetaxel**; PARP monotherapy if HRR/BRCA (olaparib, rucaparib); **Lu-PSMA** if PSMA+ (pre-taxane PSMAfore, post-taxane VISION); then **cabazitaxel**
- **ARPI-naive:** ARPI, adding a PARPi by genotype — BRCA: olaparib or niraparib + abiraterone; HRR: talazoparib + enzalutamide
- **Symptomatic bone-only, no visceral disease:** radium-223 · **MSI-H/dMMR or TMB ≥10:** pembrolizumab · asymptomatic, low burden: sipuleucel-T

Prostate cancer

HORMONAL THERAPY DOSING

Agent	Dose	Notes
Leuprolide	7.5 / 22.5 / 30 / 45 mg IM/SQ q4 / 12 / 16 / 24 wk	LHRH agonist — **flare**
Goserelin · triptorelin	3.6 mg SQ q4wk or 10.8 q12wk · 3.75 q4wk / 11.25 q12wk	Agonists
Degarelix	240 mg SQ load → 80 mg SQ q4wk	Antagonist, no flare; injection-site reactions
Relugolix (HERO)	**360 mg PO load → 120 mg PO daily**	Oral antagonist: sustained castration 97% vs 89%, castrate day 4 56% vs 0%, MACE 2.9% vs 6.2%. Adherence matters — T recovers fast
Transdermal estradiol (PATCH)	4 × 100 µg patches twice weekly → 3 once castrate	Not licensed for prostate cancer in the US
Bicalutamide	50 mg daily (flare cover; with ADT) · 150 mg (RTOG 9601)	First-gen antiandrogen
Abiraterone	**1000 mg fasting + prednisone 5 mg daily (mHSPC) or 5 mg BID (mCRPC)**	**Low-dose option: 250 mg with a low-fat meal** — non-inferior PSA decline (phase 2) at a quarter of the dose
Enzalutamide	160 mg daily	CYP3A4/2C9/2C19 inducer
Apalutamide	240 mg daily	Rash, hypothyroidism
Darolutamide	600 mg BID with food	Least CNS penetration
Abiraterone + dexamethasone	1000 mg + dexamethasone 0.5 mg daily	Steroid switch after abi/pred progression (modest PSA responses)

Abiraterone 1000 mg must be taken fasting — food raises exposure several-fold, which is the basis of the low-dose-with-food option.

SPECIAL SITUATIONS

- **ADT continues indefinitely** in CRPC — the AR axis is still active
- **LHRH agonist flare:** if cord/obstructive risk, bicalutamide ~7 days before the agonist, or use an antagonist (castrate by day 3–4)
- **New metastatic disease with cord compression or impending obstruction:** dexamethasone + RT/surgery, and start castration with a **GnRH antagonist** (degarelix) or orchiectomy — never an agonist alone
- Rapid progression with **disproportionately low PSA**, visceral/lytic mets → suspect **neuroendocrine transformation**; biopsy, treat with platinum + etoposide
- **Check testosterone** on ADT when PSA rises — not castrate means the ADT is failing, not the cancer

CHEMO, TARGETED, RADIOLIGAND DOSING

Regimen	Dose
Docetaxel (mHSPC)	**75 mg/m² q3wk ×6, no prednisone required** (CHAARTED; PEACE-1/ARASENS triplet)
Docetaxel (mCRPC)	75 mg/m² q3wk + prednisone 5 mg BID; 50 mg/m² q2wk if frail
Cabazitaxel	**20 mg/m²** q3wk (PROSELICA non-inferior) or 25 mg/m² + prednisone; **G-CSF prophylaxis**
Olaparib	300 mg BID ± abiraterone 1000 mg + prednisone 5 mg BID
Niraparib / abiraterone	**200** / **1000 mg** fixed-dose tablet daily + prednisone **10 mg (mCRPC) or 5 mg (mCSPC)**
Talazoparib	**0.5 mg daily** + enzalutamide 160 mg
Rucaparib	600 mg BID
Capivasertib	**400 mg BID, 4 days on / 3 off** + abiraterone 1000 mg + prednisone
Lu-177-PSMA-617	**7.4 GBq IV q6wk ×up to 6** — requires PSMA PET positivity
Radium-223	**55 kBq/kg IV q4wk ×6** (= the trials' 50 kBq/kg, re-standardized)
Pembrolizumab	200 mg q3wk — MSI-H/dMMR or TMB ≥10 (tumor-agnostic)
Sipuleucel-T	3 infusions q2wk — asymptomatic / minimally symptomatic mCRPC

BONE HEALTH & SUPPORTIVE CARE

- **ADT bone loss:** denosumab **60 mg SQ q6 mo** or zoledronic acid; calcium + vitamin D
- **CRPC bone metastases:** denosumab **120 mg SQ q4wk** or zoledronic acid **4 mg IV** q3–4 wk (q12wk acceptable) to prevent skeletal events. **No bone agent is indicated for SRE prevention in mHSPC**
- **Before bone agents:** dental evaluation (osteonecrosis of the jaw); correct calcium and vitamin D — denosumab causes hypocalcemia; renally dose zoledronic acid
- **Hot flashes:** venlafaxine, gabapentin, or medroxyprogesterone. **Exercise** (resistance + aerobic) counters fatigue and sarcopenia

MONITORING & TOXICITY

- **ADT (all):** baseline **DEXA**, calcium + vitamin D, bone agent if osteoporotic; **lipids, A1c, BP** (metabolic syndrome, CV risk); hot flashes (venlafaxine, gabapentin); fatigue, sarcopenia → resistance exercise; anemia; cognition
- **ARPIs: falls and fractures**; fatigue; hypertension. **Enzalutamide/apalutamide: seizure risk** — avoid with seizure history. Apalutamide: rash, TSH. Check **drug interactions** (enza/apa are strong inducers — DOACs, statins, immunosuppressants)
- **Abiraterone:** mineralocorticoid excess — **hypokalemia, hypertension, edema**; **LFTs q2wk ×3 mo then monthly**; take 1000 mg on an empty stomach
- **PARPi: anemia** (talazoparib grade ≥3 43–49%) — CBC monthly, transfuse / dose-reduce; MDS/AML rare; **VTE with olaparib + abiraterone** (7% vs 2%)
- **Lu-PSMA: xerostomia** (~45%), nausea, fatigue, **cytopenias** (CBC before each dose), renal function
- **Radium-223:** ANC ≥1.5, platelets ≥100 before first dose; **not with abiraterone**; bone agent if combined with enzalutamide
- **Capivasertib:** diarrhea, **hyperglycemia** (fasting glucose/A1c), rash
- **Taxanes:** neutropenia, neuropathy (docetaxel), diarrhea (cabazitaxel)

KEY TRIALS

EMBARK	Enza + leuprolide in high-risk BCR — MFS and **OS HR 0.60**; monotherapy no OS gain
CHAARTED / LATITUDE	Docetaxel (high volume) / abiraterone (high risk) with ADT in mHSPC
PEACE-1 / ARASENS	Triplet: ADT + docetaxel + abiraterone / darolutamide
SPARTAN / PROSPER / ARAMIS	ARPI in nmCRPC — MFS ~2 yr longer, OS HR 0.69–0.78
TALAPRO-2	Talazoparib + enza 1L mCRPC — OS HR 0.80; label HRR-only
VISION / PSMAfore	Lu-PSMA post-taxane (OS HR 0.62) / pre-taxane (rPFS only)
CARD	Cabazitaxel beats a second ARPI — OS HR 0.64
ERA 223 / PEACE-3	Radium + abiraterone harms; + enzalutamide helps with bone protection
HERO	Oral relugolix vs leuprolide — better castration, fewer MACE
AMPLITUDE / CAPItello-281 / PSMAddition	Biomarker-directed mHSPC add-ons, all approved 2025–26